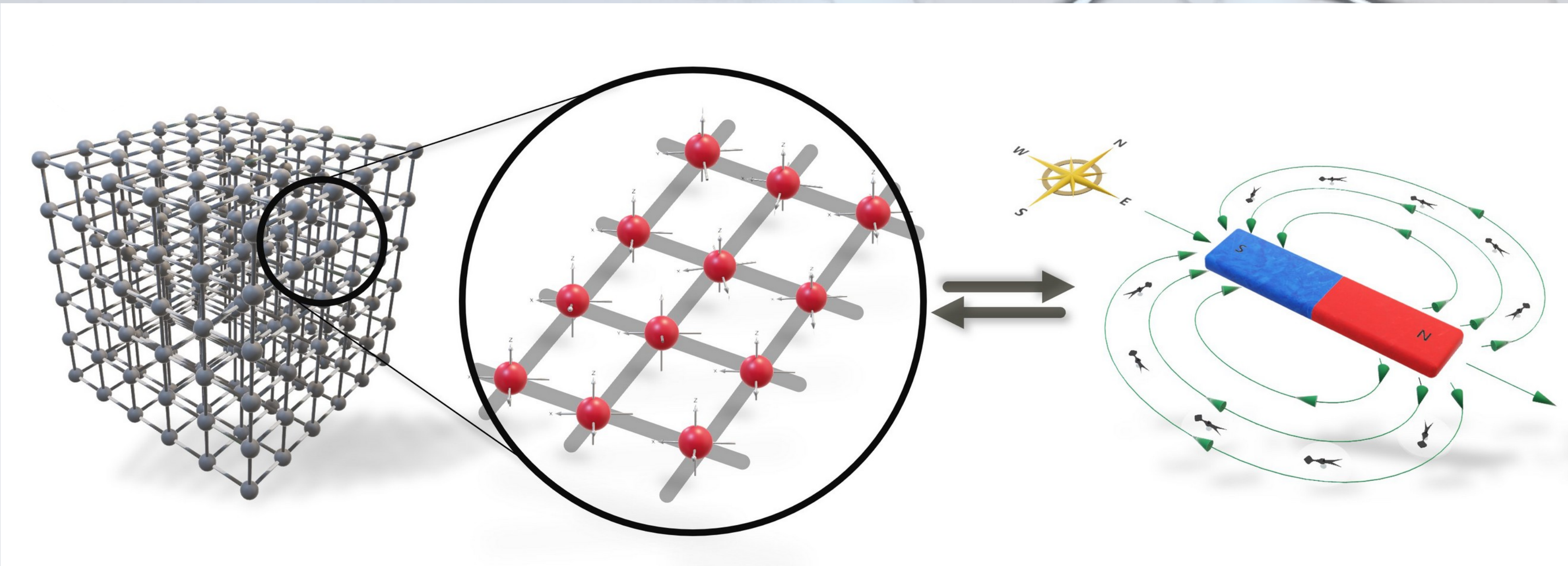
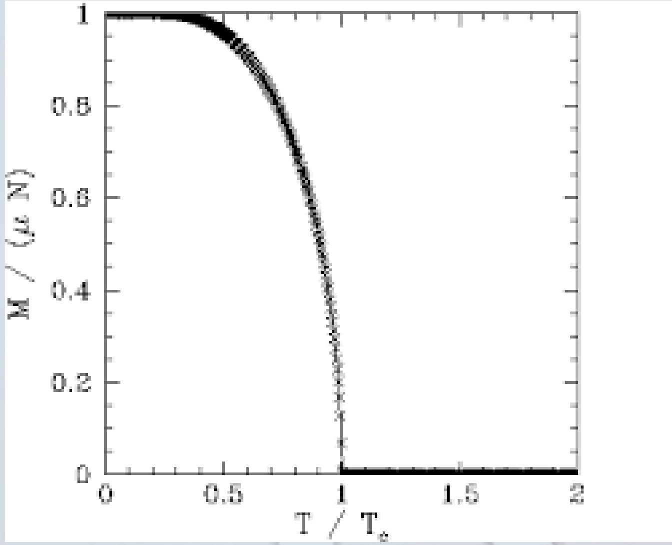
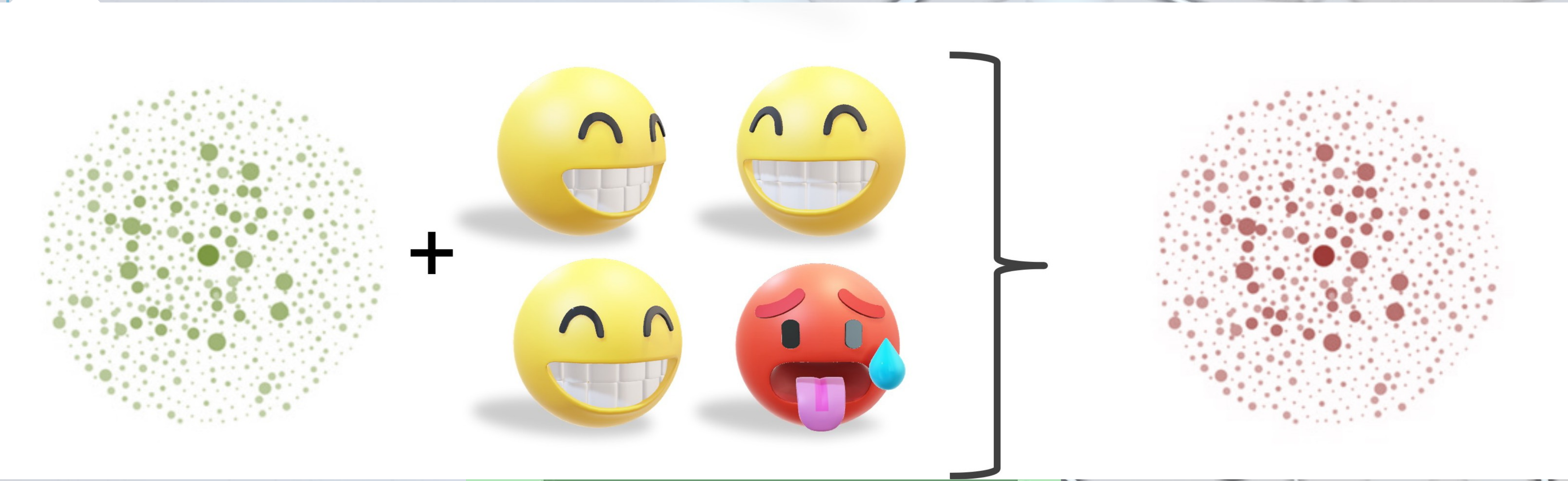
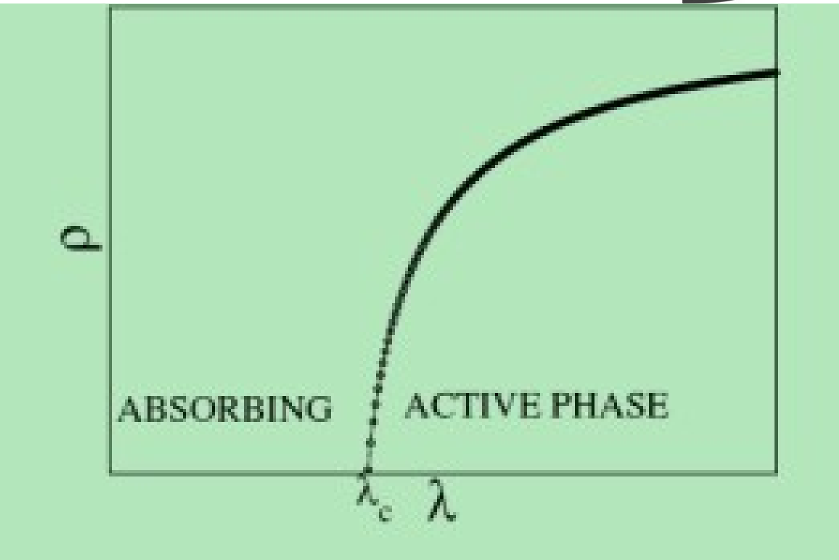
Phase Transition





μ is the atomic *magnetic moment*. *M* is the net magnetization




ρ
ABSORBING
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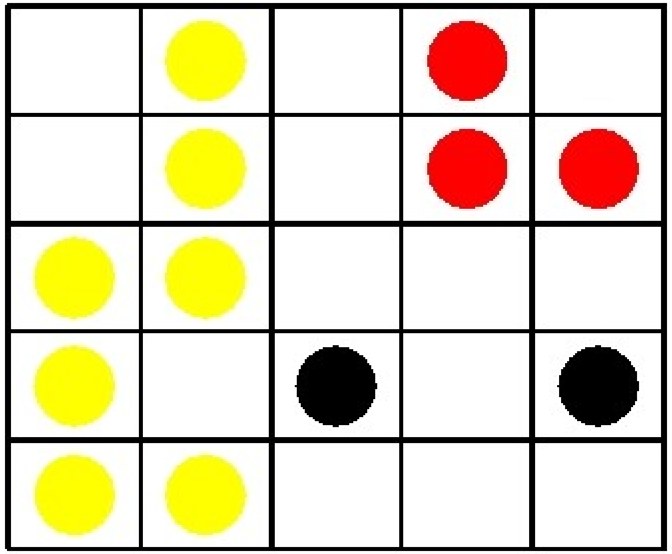
Percolation Theory

Percolation Theory



Percolation in 2d square lattice of linear size L = 5. Sites are occupied with probability p. In the lattice above, we have one cluster of size 7, a cluster of size 3 and two clusters of size 1 (isolated sites).

- Let each site in a lattice be occupied at random with probability p, that is, each site is occupied (with probability p) or empty (with probability $1-p$) independent of the status (empty or occupied) of any of the other sites in the lattice. We call p the occupation probability or the concentration.

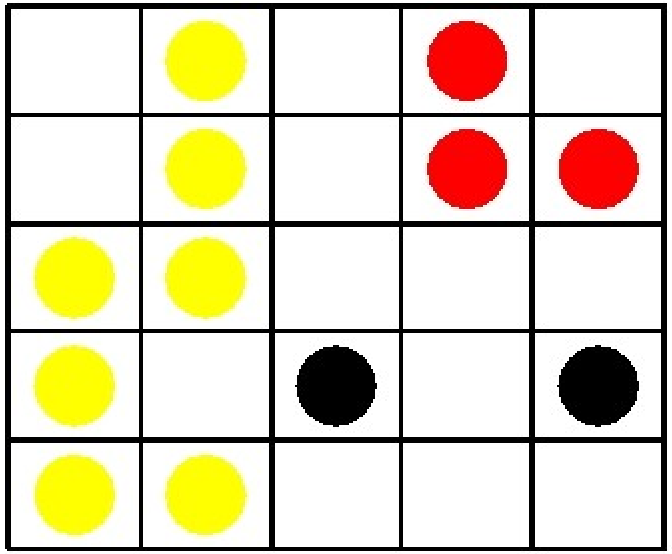
- A cluster is a group of nearest neighbouring occupied sites.

- The cluster number $n_s(p)$ denotes the number of s-clusters per lattice site.

- The percolation threshold p_c is the concentration (occupation probability) p at which an *infinite cluster* appears for the first time in an infinite lattice.

Dr. Kim Christensen

Percolation in 1-d



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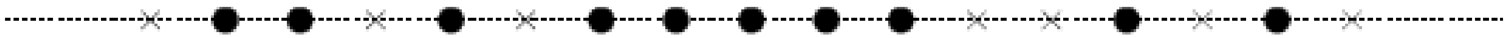
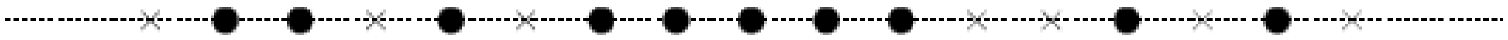


Sites are occupied with probability p. The crosses are empty sites, the solid circles are occupied sites. In the part of the infinite 1d lattice shown above, there is one cluster of size 5, one cluster of size 2, and three clusters of size 1.

Dr. Kim Christensen

Percolation in 1-d

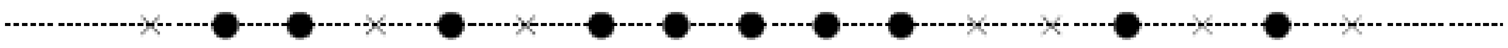


- Focus: Critical occupation probability at which an *infinite cluster* is obtained for the first time.

- Clearly, in 1d this can only be achieved if all sites are occupied, that is, the percolation threshold $p_c = 1$, as a single empty site would prevent a cluster to percolate.

- Let $\Pi(p; L)$ denote the probability that a lattice of linear size L percolates at concentration p.

Percolation in 1-d



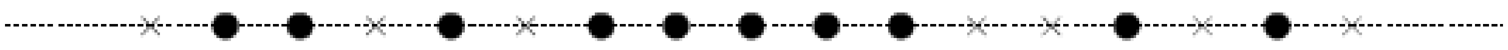
- Focus: Critical occupation probability at which an infinite cluster is obtained for the first time.
- Clearly, in 1d this can only be achieved if all sites are occupied, that is, the percolation threshold $p_c = 1$, as a single empty site would prevent a cluster to percolate.
- Let $\Pi(p; L)$ denote the probability that a lattice of linear size L percolates at concentration p.
- As the events of occupying sites are independent, all sites are occupied with probability $\Pi(p; L) = p^L$
  Thus

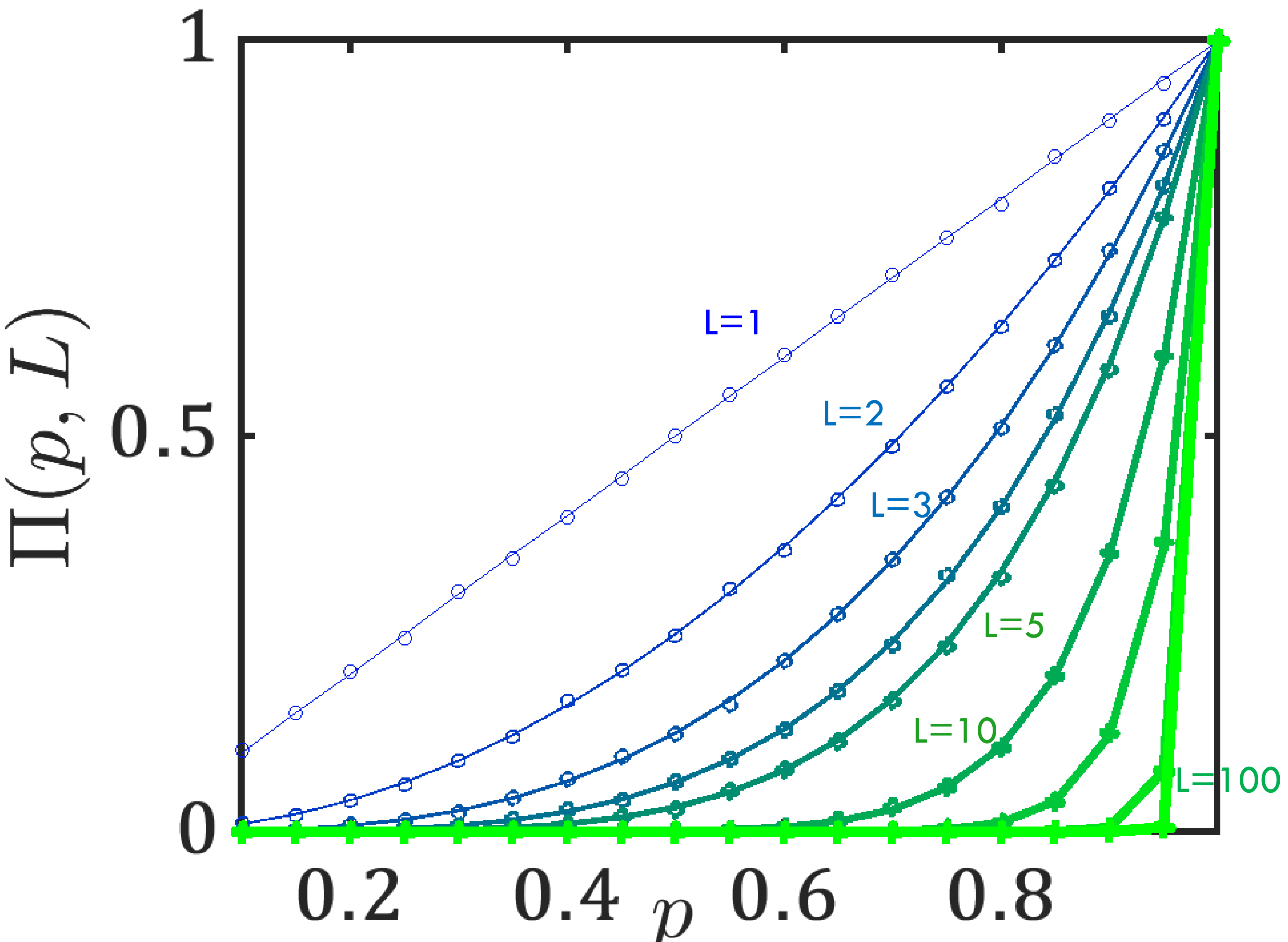
$$\lim_{L\to\infty} \Pi(p; L) = \lim_{L\to\infty} p^L = \begin{cases} 0 \text{ for } \mathrm{p} < 1 \\ 1 \text{ for } \mathrm{p} = 1 \end{cases}$$

Show it numerically

Dr. Kim Christensen

Percolation in 1-d





Percolation in 1-d

Question: If the probability of one site being occupied is p, what is probability of two sites next to each other being occupied?

Since the probability of each site being occupied is independent, it is just p^2.

This generalizes for the probability that s specific sites are occupied to p^s. The probability that a sites neighbor is empty is $(1-p)$.

- Combining the above results, the probability a site is the left edge of a cluster of length s is $p^s(1-p)^2$.

- If we have a long chain of length L, how many clusters of length 5 will there be? Well, if we assume the chain is very long, we can ignore the effects of the end of chain. Then the probability of each site being the left edge of a cluster of length 5 is $p^5(1-p)^2$ and can be at any one of the L sites.

Percolation in 1-d

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- Thus the total number of clusters of size 5 is $p^5(1-p)^2 \times L$. It is normally more practical to talk about the number of clusters per lattice site, which is the above expression divided by L, or just $p^5(1-p)^2$.

- Thus, for clusters of s sites, we define n_s, normalized cluster, as the number of clusters with size s per lattice site.

Percolation in 1-d

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- Thus, for clusters of s sites, we define n_s, normalized cluster, as the number of clusters with size s per lattice site.

- The probability that a site is a part of a cluster of size s is **actually** higher than the normalized cluster number, because it does not have to be at the left end. It can be at any of the s sites. Therefore, the probability a single site is in a cluster of size s is $n_s \times s$.

- Since all sites have equal probability of being occupied (or empty), the probability that an arbitrary site is part of an s-cluster is s times the probability of it being the LHS of the cluster.

Percolation in 1-d

$$\begin{aligned} n_s(p) &= (1-p)^2 p^s \\ &= (1-p)^2 \exp(\ln(p^s)) \\ &= (1-p)^2 \exp(s\ln(p)) \\ &= (p_c-p)^2 \exp(-\frac{s}{s_\xi}) \end{aligned}$$

with the definition of a *cutoff cluster size* or *characteristic cluster size*

$$s_\xi = \frac{-1}{\ln(p)} = \frac{-1}{\ln(p_c-(p_c-p))} \to \frac{1}{p_c-p} = (p_c-p)^{-1} \qquad \text{for } p \to p_c,$$

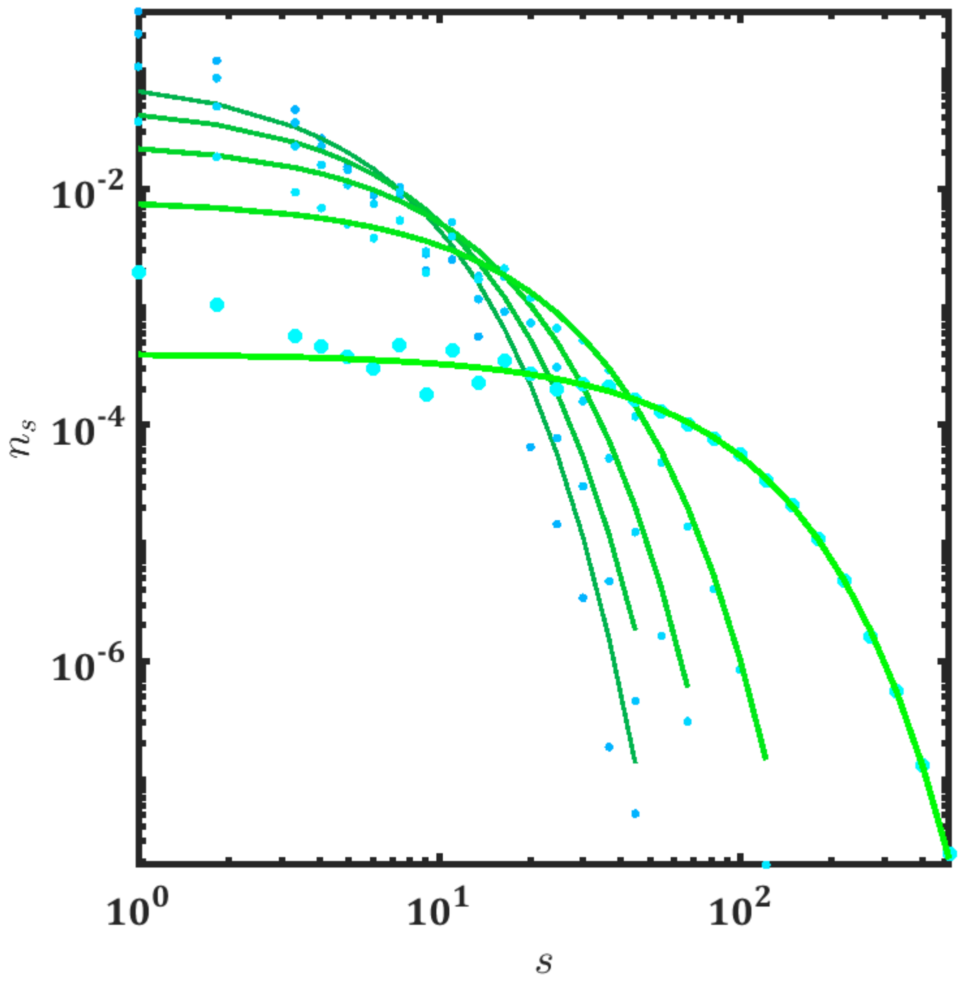
where we, to obtain the limit, have used $p_c = 1$ and the Taylor expansion

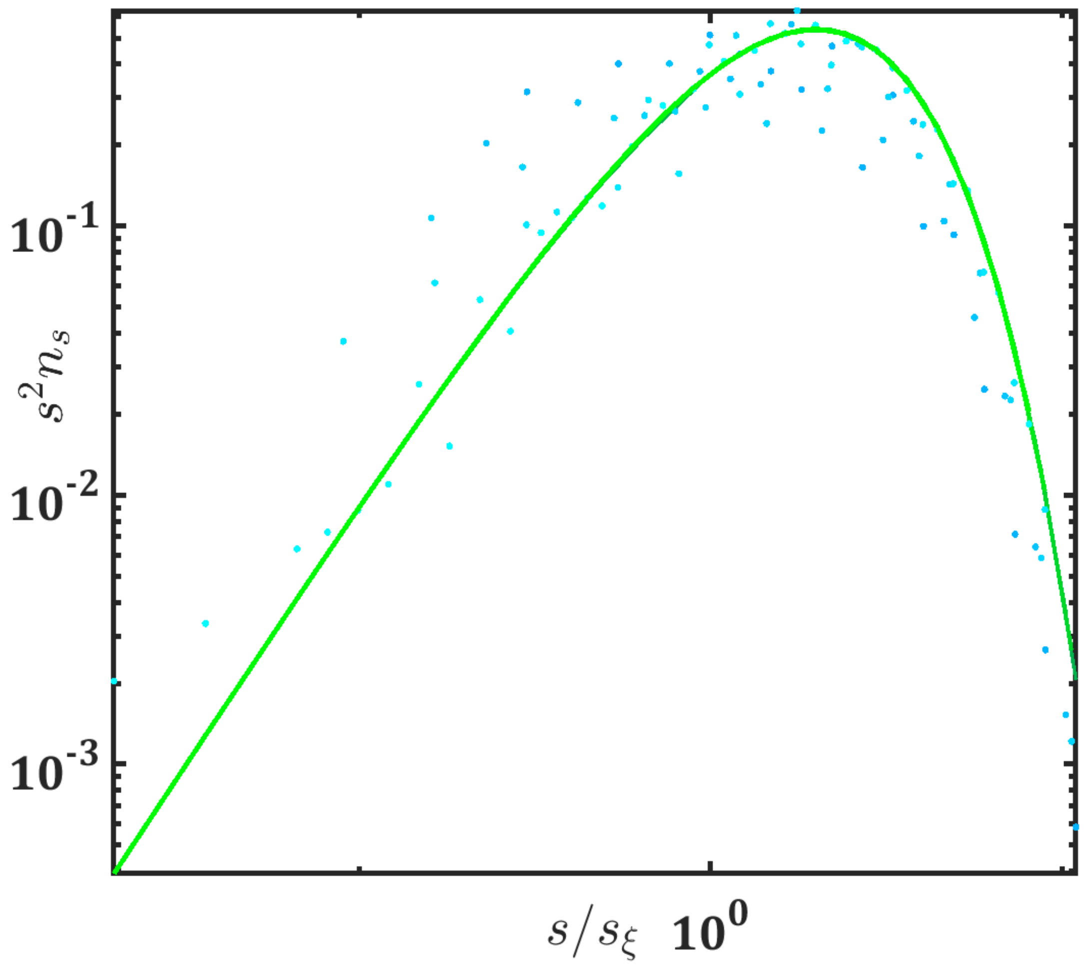
$$\ln(1-x) = -x - \frac{1}{2}x^2 - \frac{1}{3}x^3 - \cdots \approx -x$$

where the last approximation is valid for $x \to 0$, see Fig. 1.5.

Q: Show that (in 1-d lattice), normalized cluster defined as the number of clusters with size *s* per lattice site can be written as n_s(p)=(1-p)^2 p^s. Prove that cut-off cluster size is s_\xi=(p_c-p)^-1. Here the probability of one site being occupied is *p*

Percolation in 1-d





Percolation in 1-d

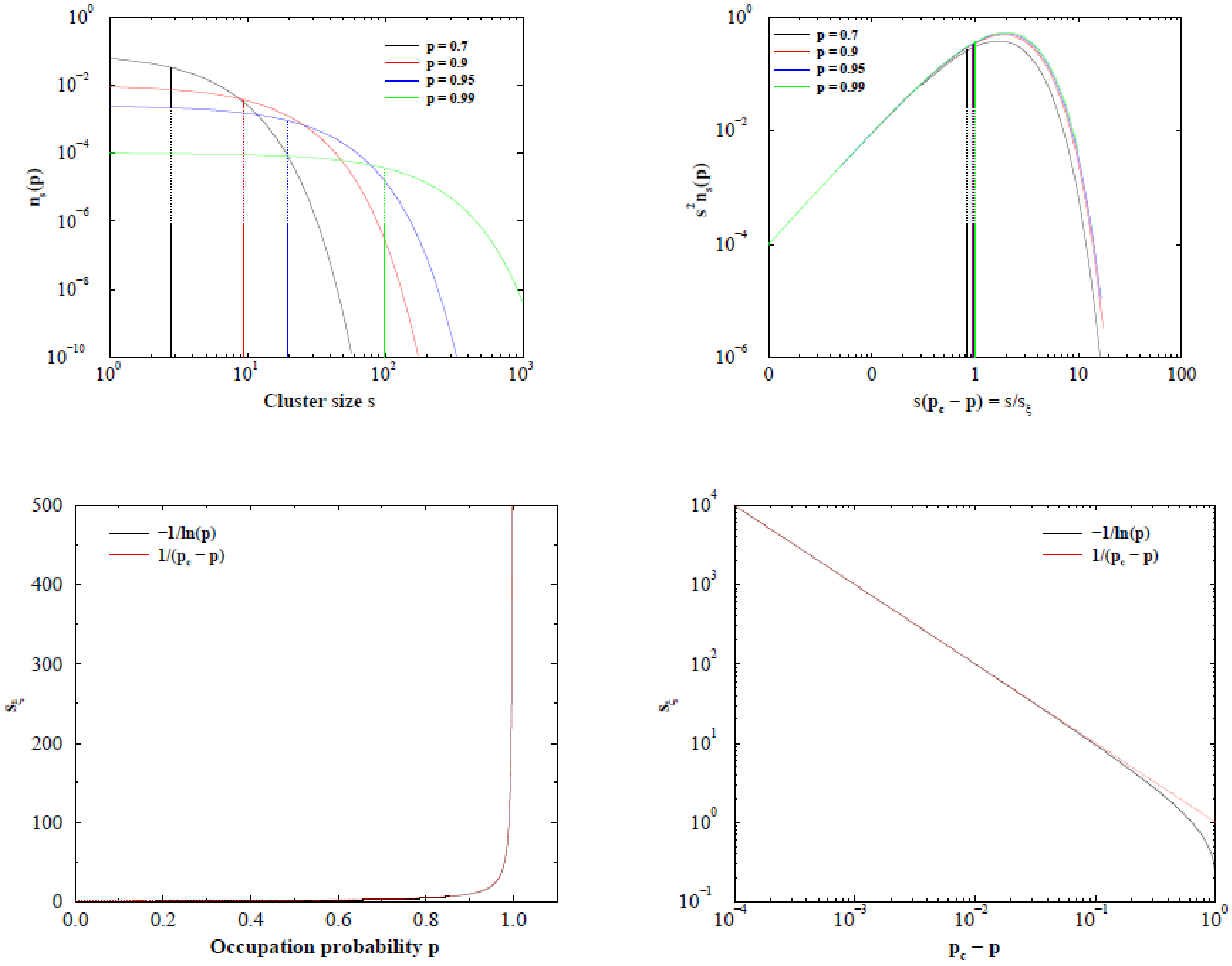


Figure 1.5: Percolation in 1d. (a) The cluster number distribution $n_s(p) = (p_c - p)^2 \exp(-\frac{s}{s_\xi})$ for various values of p approaching $p_c = 1$. The vertical lines indicate the cutoff cluster size $s_\xi(p)$. (b) By plotting $s^2 n_s(p)$ versus $s/s_\xi \approx s(p_c - p)$, all the data collapses onto a function $x^2 \exp(-x)$. (c) The characteristic cluster size $s_\xi = -\frac{1}{ln(p)}$ diverges when $p \to p_c = 1$. (d) In the limit $p \to p_c = 1$, $s_\xi \propto (p_c - p)^{-1}$.

Percolation in 1-d

Let us continue our journey into $1d$ percolation theory. For $p < p_c$ we can state that the probability that an arbitrary site belongs to **any** (finite) cluster is simply the probability p of it being occupied. Since the probability that an arbitrary sites belongs to an s-cluster is given by $sn_s(p)$, we arrive at

$$\sum_{s=1}^{\infty} sn_s(p) = p \qquad \text{for } p < p_c.$$

$$\begin{aligned}
\sum_{s=1}^{\infty} sn_s(p) &= \sum_{s=1}^{\infty} s(1-p)^2 p^s \\
&= (1-p)^2 \sum_{s=1}^{\infty} p \frac{d(p^s)}{dp} \\
&= (1-p)^2 p \frac{d}{dp}\left(\sum_{s=1}^{\infty} p^s\right) \\
&= (1-p)^2 p \frac{d}{dp}\left(\frac{p}{1-p}\right) \\
&= p.
\end{aligned}$$

Q: Show that (in 1-d lattice), normalized cluster defined as the number of clusters with size *s* per lattice site can be written as n_s(p)=(1-p)^2 p^s. Prove that cut-off cluster size is s_\xi=(p_c-p)^-1. Show that sum(s n_s)=p
Here the probability of one site being occupied is *p*.

Percolation in 1-d

How large on average is a cluster or, equivalently, how large is a cluster on average to which an occupied site belongs?

The probability that a site is occupied is p. The probability that an arbitrary site belongs to an s-cluster is $sn_s(p)$. Define anew variable w_s

$$w_s = \frac{sn_s(p)}{p} = \frac{sn_s(p)}{\sum_{s=1}^{\infty} sn_s(p)}.$$

Which is in some sense a "weighting" of relative cluster sizes.

Percolation in 1-d

Thus, the *mean cluster size* or *average cluster size* $S(p)$ is given by

$$\begin{aligned} S(p) &= \sum_{s=1}^{\infty} s w_s \\ &= \sum_{s=1}^{\infty} \frac{s^2 n_s(p)}{\sum_{s=1}^{\infty} n_s(p) s} \\ &= \frac{1}{p}(1-p)^2 \sum_{s=1}^{\infty} s^2 p^s \\ &= \frac{1}{p}(1-p)^2 \left(p\frac{d}{dp}\right)^2 \left(\sum_{s=1}^{\infty} p^s\right), \end{aligned}$$

where the operator

$$\left(p\frac{d}{dp}\right)^2 = \left(p\frac{d}{dp}\right)\left(p\frac{d}{dp}\right) \neq p^2\frac{d^2}{dp^2}.$$

Using the formula for summing a geometric series and the operator $p\frac{d}{dp}$ twice, we finally arrive at

$$S(p) = \frac{1+p}{1-p} = \frac{p_c+p}{p_c-p}$$

where the last equality follows because in $1d$ the critical occupation probability $p_c = 1$.

Percolation in 1-d

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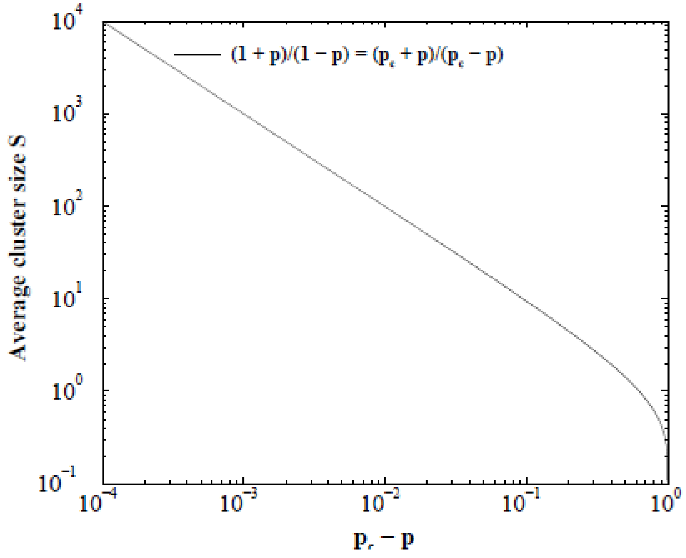
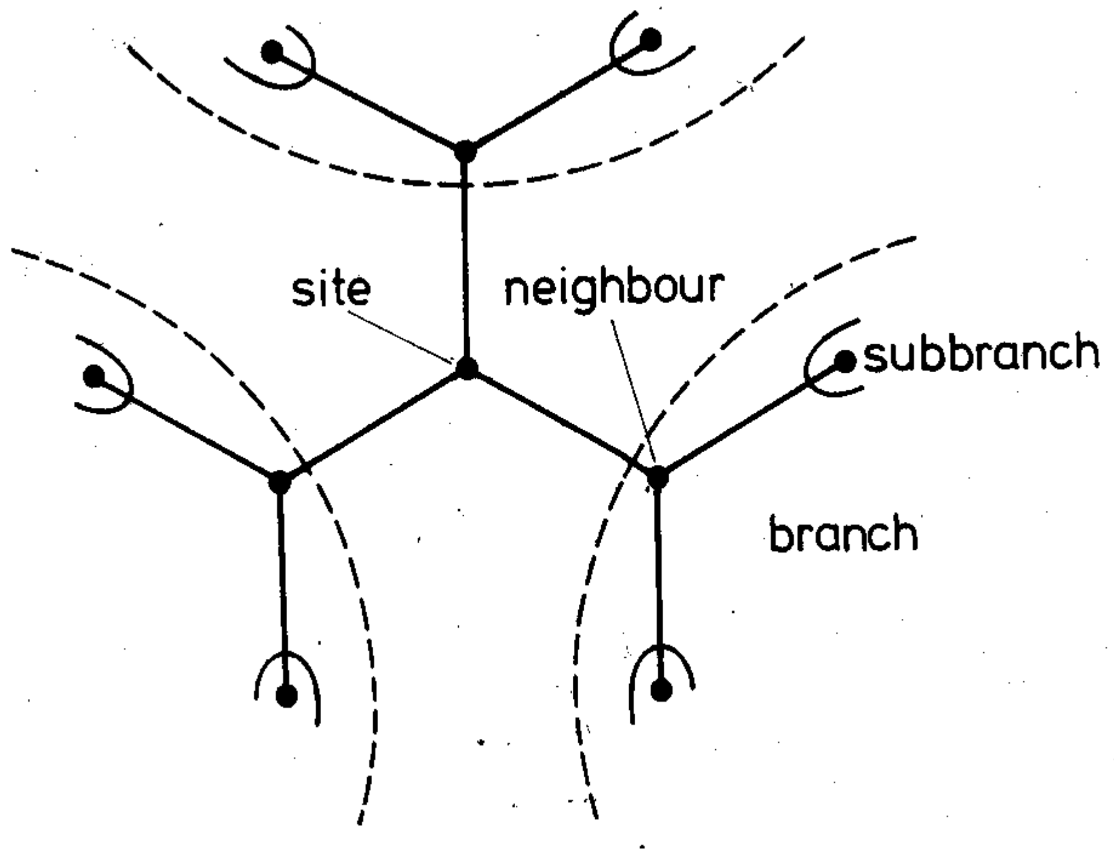
In order to investigate in detail how the mean cluster size diverges, when taking the limit $p \to p_c^-$, we note that the numerator in Eq.(1.6) approaches $2p_c$, so

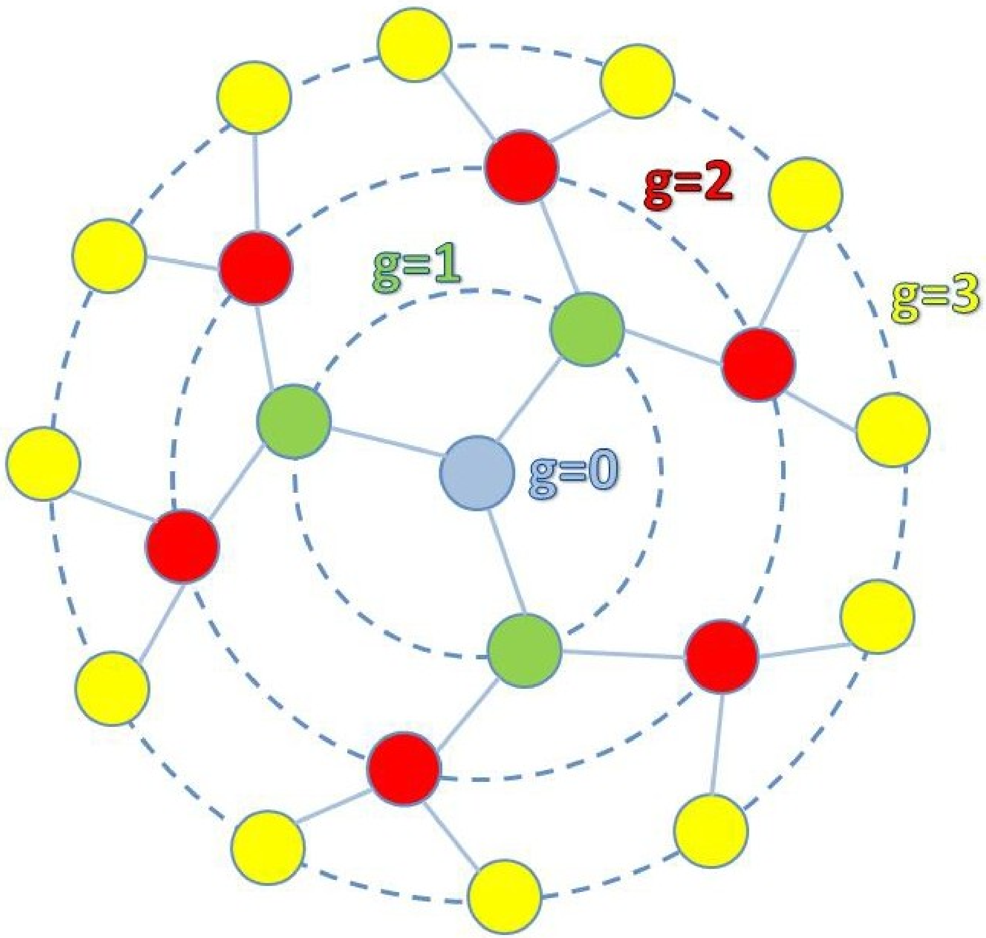
$$S(p) = \frac{p_c + p}{p_c - p} \to \frac{2p_c}{p_c - p} \propto (p_c - p)^{-1} \qquad \text{for } p \to p_c^- .$$

Thus in $1d$, the mean cluster diverges like a power law in the quantity $(p_c -$



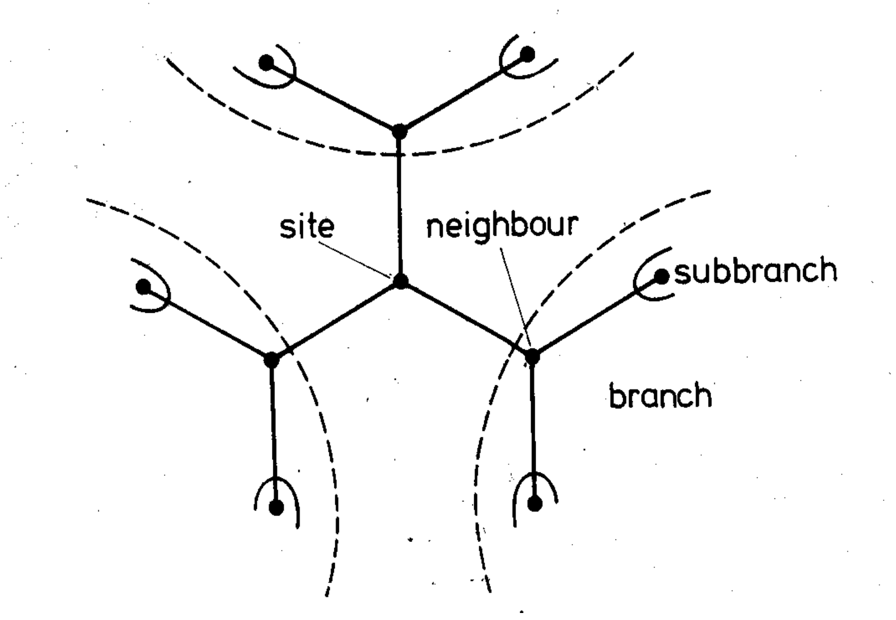
Percolation in Bethe Lattice





Each site has z neighbouring sites
Each branch gives rise to $z-1$ other branches
There are no closed loops
Z=3;

Percolation in Bethe Lattice



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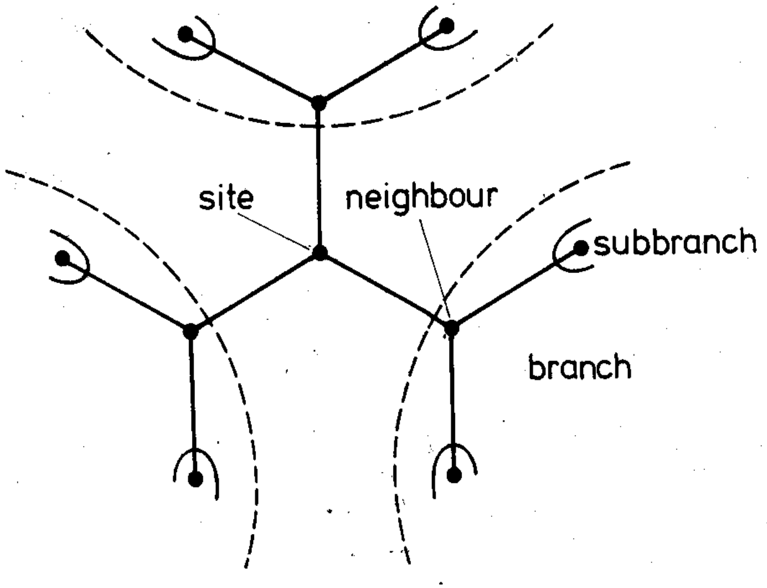
Percolation in Bethe Lattice



Q. Prove that, a) the number of surface sites relative to the total number of sites approaches a constant.

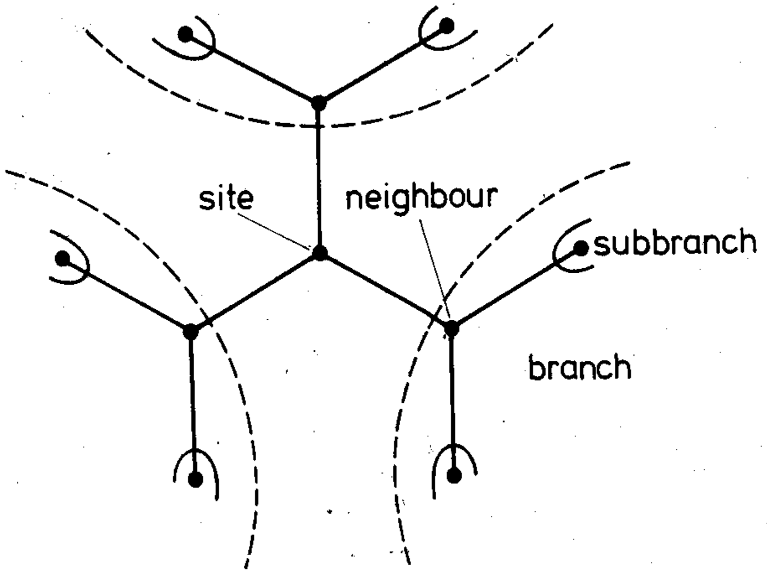
- Let g denote the generation, that is, the distance from a "centre site".
- Note, however, that in an infinite Bethe lattice, all sites are equivalent, so the notion of a "centre site" is not to be taken literally.

- In the figure left, the " first ring" of three sites belong to generation g = 1, the second "ring" of six sites belongs to the second generation and so on.

The total number of sites in a Bethe lattice consisting of g generations is

$$\begin{aligned}\text{Total no. sites} &= 1 + 3\cdot(1+2+\cdots+2^{g-1}) \\ &= 1 + 3\cdot\frac{1-2^g}{1-2} \\ &= 3\cdot 2^g - 2,\end{aligned}$$

Percolation in Bethe Lattice



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$$\begin{aligned} \text{Total no. sites} &= 1 + 3 \cdot (1 + 2 + \cdots + 2^{g-1}) \\ &= 1 + 3 \cdot \frac{1 - 2^g}{1 - 2} \\ &= 3 \cdot 2^g - 2, \end{aligned}$$

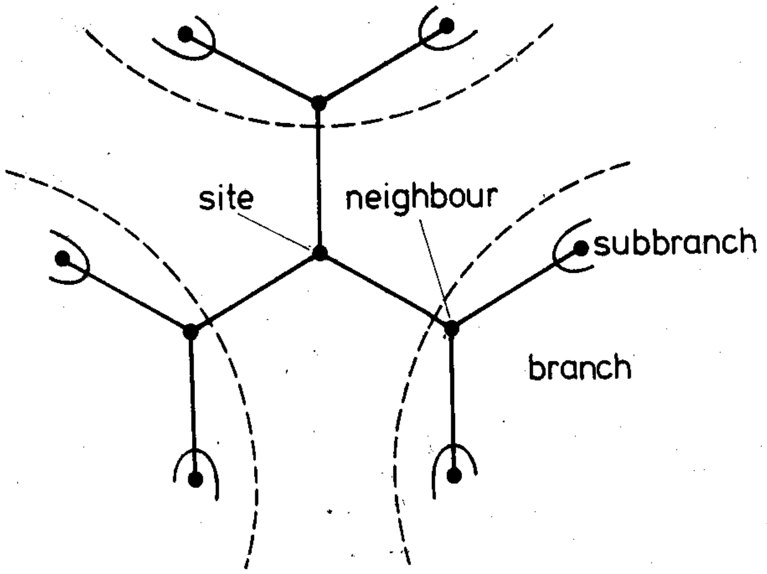
while the number of surface sites is $3 \cdot 2^{g-1}$. Thus

$$\frac{\text{No. of surface sites}}{\text{Total no. of sites}} = \frac{3 \cdot 2^{g-1}}{3 \cdot 2^g - 2} \to \frac{1}{2} \qquad \text{for } g \to \infty.$$

and the surface/volume tends to a constant.

Notice, that for $z = 2$, the $1d$ percolation problem, we recover $p_c = 1$. For $z = 3$, e.g., $p_c = \frac{1}{2}$. Note that in a hypercubic lattice with $d \to \infty$ one would expect $p_c \to 1/(2d - 1)$ in order to be able to continue walking along a path of occupied sites.

Percolation in Bethe Lattice



What is the critical occupation probability in a Bethe lattice, that is, at which occupation probability does an infinite cluster (path) appear for the first time in an infinite Bethe lattice? Starting from a "centre site" and going outwards, we encounter $(z-1)$ new neighbours. Thus, on average, we have $p(z-1)$ *new* occupied sites on which we can continue the path. The critical occupation probability is determined by the equation

$$p_c(z-1) = 1 \Leftrightarrow p_c = \frac{1}{z-1}.$$

Percolation in Bethe Lattice

The strength of the infinite cluster $P(p)$ is the probability that an arbitrary site belongs to the infinite cluster

Let define the quantity Q as the probability that an arbitrary site is NOT connected to infinity through a fixed branch originating at this site.

Percolation in Bethe Lattice

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Thus, the strength $P(p)$ of the infinite network, that is, the probability that an arbitrarily selected site is connected to infinity by occupied sites is

$P(p) =$ (Prob. site is occupied) ×(Prob. at least ONE branch lead to infinity) = $p \times (1 - Q^3)$

Percolation in Bethe Lattice

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$$= (1 - p) + pQ^2$$

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$$= (1-p) + pQ^2$$

It is quadratic, and the solution of Q will be

$$Q = \frac{1 \pm \sqrt{(2p-1)^2}}{2p} = \begin{cases} 1 \\ \frac{1-p}{p}. \end{cases}$$

Percolation in Bethe Lattice

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Using a Taylor expansion for $P(p)$ around $p = p_c = \frac{1}{2}$, it can be shown that

$$P(p) \propto (p - p_c) \qquad \text{for } p \to p_c^+.$$

Percolation Theory (in 2-d)

- Percolation theory is a branch of statistical physics that deals with the behavior of connected clusters in random systems. The basic idea is to study how connectivity emerges in a disordered medium. The model involves a **lattice or network of sites**, and each site is occupied with a certain probability (p) or vacant with the complementary probability $(1-p)$. p is called as occupation probability.

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- The most common example is the percolation on a square lattice, where sites are connected to their nearest neighbors. ***If a cluster of connected sites spans the lattice from one side to another, it is said to percolate.*** The percolation phase transition occurs at a critical probability (p_c), separating a regime where isolated clusters dominate from a regime where a **giant percolating cluster emerges.**

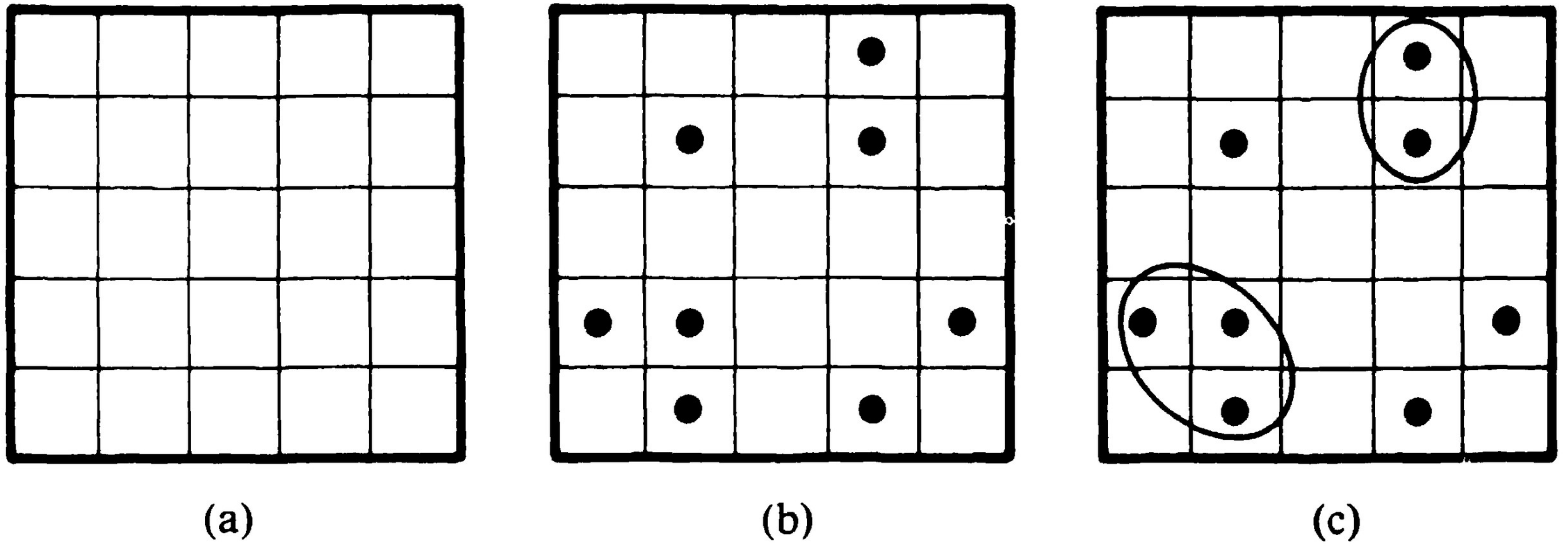
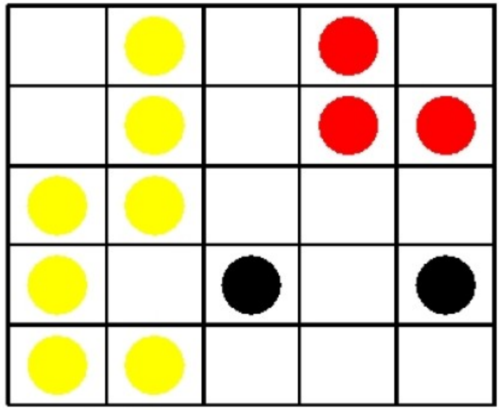


Fig. 1. Definition of percolation and its clusters: (a) shows parts of a square lattice; in (b) some squares are occupied with big dots; in (c) the 'clusters', groups of neighbouring occupied squares, are encircled except when the 'cluster' consists of a single square.

Dietrich Stauffer
Amnon Aharony

Percolation Theory (contd.)



Percolation in $2d$ square lattice of linear size $L = 5$. Sites are occupied with probability p. In the lattice above, we have one cluster of size 7, a cluster of size 3 and two clusters of size 1 (isolated sites).

A cluster is a group of nearest neighbouring occupied sites.